Answer: C

11 $P_{output} = F_{resistive}v = (400)(20) = 8000\ \text{W}$

$$\eta = \frac{P_{output}}{P_{input}} = 0.16$$

$$\Rightarrow P_{input} = \frac{8000}{0.16} = 50\ \text{kW}$$

$$\text{total time taken } t = \frac{1000\ \text{m}}{20\ \text{m s}^{-1}} = 50\ \text{s}$$

$$\therefore \text{total energy required} = 50\ \text{kW} \times 50\ \text{s} = 2.5\ \text{MJ}$$

$$\Rightarrow \text{total fuel required} = \frac{2.5}{48} \times 1\ \text{kg} = 52\ \text{g}$$

Answer: C

12 Taking that the period of rotation of the Earth about its own axis is 24 hours